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Faculty of Engineering
Electronics and Communications Engineering Department

GRADUATION PROJECT THESIS — PART II

Machine Learning Techniques for PAPR Reduction in Multicarrier Communication Systems

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Declaration of Authorship

We, Nour-Eldin Mohamed Mostafa, Youssef Samir Sleem Ahmed, Muhammad Mahmoud Muhammad, Reem Nader Saeed, Basel Mohamed Ramadan, Ahmed Walid Hassan, Yosef Ahmed Mohamed, declare that this thesis titled “Machine Learning Techniques for PAPR Reduction in Multicarrier Communication Systems” and the work presented in it are our own. We confirm that this material, submitted for assessment on the programme leading to the award of Bachelor of Engineering, is entirely our own work. We have exercised reasonable care to ensure that the work is original and does not, to the best of our knowledge, breach any law of copyright; it has not been taken from the work of others except to the extent that such work has been cited and acknowledged within the text.

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Abstract

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CHAPTER 1

OFDM System Model and Classical PAPR Reduction Techniques

This chapter presents the Orthogonal Frequency-Division Multiplexing (OFDM) signal model and defines the Peak-to-Average Power Ratio (PAPR) problem that motivates the machine-learning approaches studied in subsequent chapters. A concise survey of classical PAPR reduction techniques is provided, covering signal distortion methods and signal scrambling methods, with particular attention to Selected Mapping (SLM) as the primary baseline for the deep learning system introduced in Chapter 2.

1.1 OFDM Signal Model

Orthogonal Frequency-Division Multiplexing divides a wideband channel into N narrowband, orthogonal subcarriers. Each subcarrier carries one modulated symbol from a constellation such as QPSK or M -QAM. The modulation and demodulation operations are efficiently implemented via the Inverse Fast Fourier Transform (IFFT) and Fast Fourier Transform (FFT), respectively.

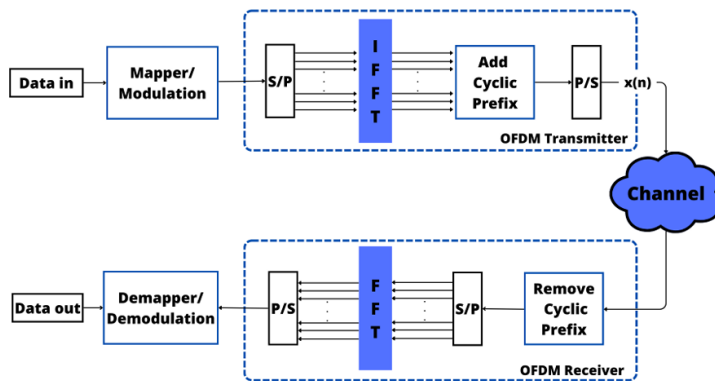


FIGURE 1.1: Block diagram of a baseband OFDM transmitter and receiver. The transmitter maps data onto N subcarriers via an N -point IFFT, adds a cyclic prefix (CP), and transmits the time-domain signal. The receiver removes the CP and applies an N -point FFT to recover the frequency-domain symbols.

1.1.1 Subcarrier Modulation via the IFFT

Let the frequency-domain data symbol vector after constellation mapping be

$$\mathbf{X} = [X_0, X_1, \dots, X_{N-1}]^T,$$

where X_k is the complex symbol transmitted on the k -th subcarrier and N is the total number of subcarriers. The time-domain OFDM signal is obtained by applying the N -point IFFT:

1.1

$$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j2\pi kn/N}, \quad n = 0, 1, \dots, N-1.$$

The $1/\sqrt{N}$ normalization preserves signal energy between the frequency and time domains (Parseval's theorem). At the receiver, the frequency-domain symbols are recovered by the complementary FFT operation:

1.2

$$X_k = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}, \quad k = 0, 1, \dots, N-1.$$

1.1.2 Cyclic Prefix

Before transmission, a **Cyclic Prefix (CP)** of length N_{cp} samples is prepended to each OFDM symbol by copying the last N_{cp} samples of the IFFT output to the front. The CP serves two purposes:

1. **ISI elimination:** provided N_{cp} exceeds the maximum channel delay spread, the CP absorbs all multipath echoes from the previous symbol.
2. **Circular convolution:** the CP converts the linear convolution with the channel into a circular convolution, enabling single-tap equalization in the frequency domain via the FFT.

1.1.3 Spectral Efficiency

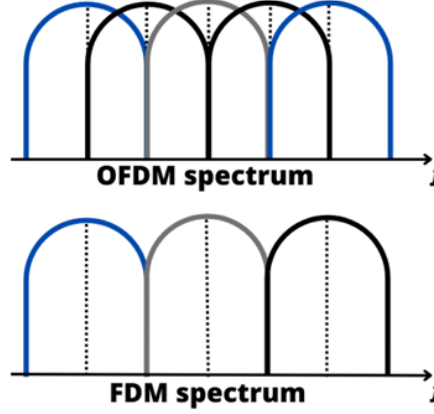


FIGURE 1.2: Spectrum comparison between conventional Frequency-Division Multiplexing (FDM) and OFDM. In FDM, guard bands between channels reduce spectral efficiency. In OFDM, orthogonal subcarriers overlap without mutual interference.

OFDM subcarriers are spaced at exactly $\Delta f = 1/T_s$, where T_s is the useful symbol duration. At every subcarrier's centre frequency, all other subcarriers have a spectral null, ensuring zero mutual interference and significantly higher spectral efficiency compared to conventional FDM.

1.2 Peak-to-Average Power Ratio

1.2.1 Definition

The time-domain OFDM signal is the superposition of N independently modulated complex exponentials. As established by the IFFT summation, each individual time-domain sample $x(n)$ is the direct sum of the contributions from all N subcarriers at that specific instant. Because the subcarriers are independent in the frequency domain, their phases are random. When these subcarrier contributions align constructively at certain time instants — for example, if all N subcarriers happen to reach their positive peak simultaneously at a specific sample n — they sum to a massive amplitude spike.

The severity of this phenomenon is quantified by the **Peak-to-Average Power Ratio (PAPR)**:

1.3

$$\text{PAPR}(x) = \frac{\max_{0 \leq n \leq N-1} |x(n)|^2}{E[|x(n)|^2]},$$

where the numerator is the peak instantaneous power (the maximum squared amplitude found among all N time samples in the given OFDM symbol) and the denominator is the average power of the OFDM symbol.

The theoretical worst-case PAPR for N subcarriers occurs when the input data causes all subcarriers to align perfectly in phase. In this scenario, the peak voltage becomes N times the voltage of a single subcarrier, yielding a peak power that is N^2 times the subcarrier power. Since the average power only scales by N , the resulting worst-case PAPR is exactly N , or $10 \log_{10}(N)$ dB.

1.2.2 CCDF Performance Metric

Since PAPR varies from one OFDM symbol to the next, it is characterised statistically using the **Complementary Cumulative Distribution Function (CCDF)**:

$$\text{CCDF}_{\text{PAPR}} = \Pr(\text{PAPR} > \text{PAPR}_0),$$

where PAPR_0 is a given threshold. For N subcarriers with independent, identically distributed symbols and Nyquist-rate sampling, the CCDF can be approximated as:

1.4

$$\Pr(\text{PAPR} \geq \text{PAPR}_0) \approx 1 - (1 - e^{-\text{PAPR}_0})^N.$$

A PAPR reduction technique shifts the CCDF curve to the left, indicating that high-PAPR events become statistically less likely.

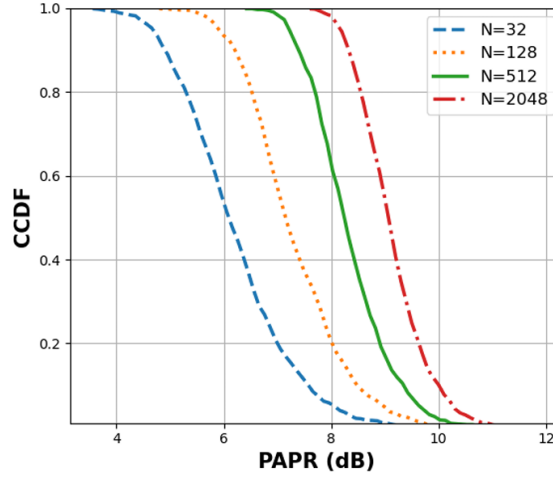


FIGURE 1.3: CCDF of PAPR for different numbers of subcarriers ($N = 32, 128, 512, 2048$) with 16-QAM modulation. As N increases, the PAPR distribution shifts to the right, confirming that more subcarriers lead to higher PAPR.

1.2.3 Power Amplifier Nonlinearity

High PAPR has direct consequences for the **High-Power Amplifier (HPA)** at the transmitter. Every power amplifier has a saturation region: when the input signal exceeds a certain amplitude, the output clips or compresses, introducing nonlinear distortion. To avoid entering this nonlinear zone, the HPA must operate with sufficient **Input Back-Off (IBO)**:

$$\text{IBO (dB)} = 10 \log_{10} \left(\frac{P_{\text{sat}}}{P_{\text{avg}}} \right),$$

where P_{sat} is the amplifier's saturation power and P_{avg} is the average input signal power. For an OFDM signal with $\text{PAPR} = 12$ dB, the IBO must be at least 12 dB, meaning the amplifier operates far below its maximum capacity. This directly reduces power efficiency and increases hardware complexity.

Reducing PAPR by even 3–4 dB translates to a significant improvement in HPA efficiency, reduced heat dissipation, and lower deployment costs, making PAPR reduction one of the most active research areas in OFDM system design.

1.3 Classical PAPR Reduction Techniques

Classical PAPR reduction techniques can be broadly classified into **signal distortion** methods, which introduce controlled modifications to the transmitted signal, and **signal scrambling** methods, which generate alternative signal representations without distortion.

1.3.1 Signal Distortion Techniques

Clipping and Filtering. The simplest approach: any signal sample exceeding a threshold A is clipped to that threshold. Filtering is subsequently applied to suppress out-of-band spectral regrowth. The clipped signal is given by:

$$x'(n) = \begin{cases} x(n), & |x(n)| \leq A, \\ \frac{A}{|x(n)|} x(n), & |x(n)| > A, \end{cases}$$

where $A = \text{CR} \cdot \sqrt{P_{\text{avg}}}$ and CR is the clipping ratio.

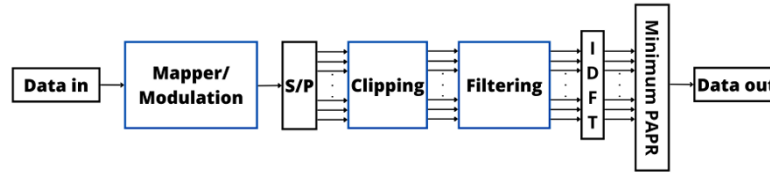


FIGURE 1.4: Block diagram of an OFDM transmitter with clipping and filtering. The signal is clipped in the time domain, then filtered in the frequency domain to remove out-of-band distortion.

While clipping and filtering is simple to implement and requires no side information, it introduces in-band distortion (increasing BER) and out-of-band radiation.

Active Constellation Extension (ACE). ACE extends outer constellation points away from the decision boundaries to reduce peak power, without affecting BER. No side information is required, as the extended points remain within their correct decision regions.

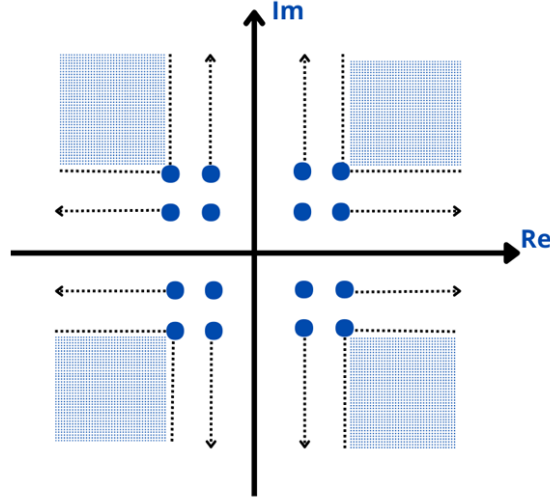


FIGURE 1.5: Active Constellation Extension for 16-QAM. Outer constellation points are extended into feasible regions (shaded), reducing the PAPR without crossing decision boundaries.

Tone Reservation (TR) and Tone Injection (TI). TR reserves a set of subcarriers exclusively for peak-cancelling signals, while TI adds carefully designed tones to the data subcarriers. Both reduce PAPR without distorting data symbols, at the cost of reduced spectral efficiency (TR) or increased average power (TI).

1.3.2 Signal Scrambling Methods

Signal scrambling methods alter the signal representation *without distortion* to find a low-PAPR version.

Partial Transmit Sequence (PTS). The input data block is partitioned into V disjoint sub-blocks, each transformed by an IDFT independently. The sub-blocks are then weighted by phase factors $b_v \in \{e^{j\phi} : \phi \in \Phi\}$ and summed. The phase combination yielding the lowest PAPR is selected for transmission.

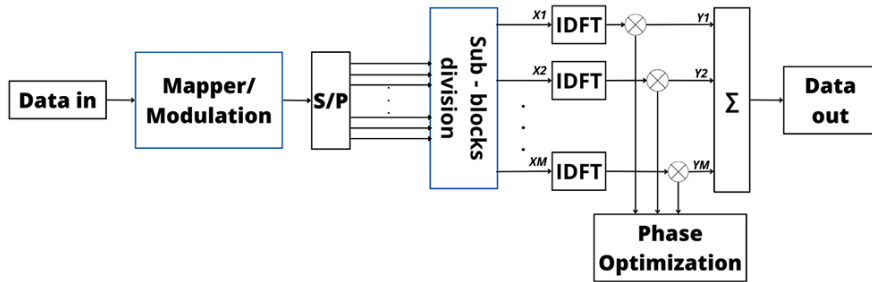


FIGURE 1.6: Block diagram of an OFDM transmitter with Partial Transmit Sequence (PTS). The data is split into V sub-blocks, each weighted by a phase factor after the IDFT. The combination with the lowest PAPR is transmitted.

Selected Mapping (SLM). Selected Mapping generates U alternative representations of the same data block by multiplying the frequency-domain symbols with U different phase sequences ψ_u , each applied element-wise (Hadamard product):

$$\mathbf{X}_u = \mathbf{X} \odot \psi_u, \quad u = 1, 2, \dots, U.$$

The corresponding time-domain candidates are obtained via the IDFT, and the candidate with the minimum PAPR is selected for transmission:

1.5

$$\hat{u} = \arg \min_{1 \leq u \leq U} \text{PAPR}(\mathbf{x}_u).$$

The receiver requires knowledge of which phase sequence was used, transmitted as $\lceil \log_2 U \rceil$ bits of **side information (SI)** per OFDM symbol.

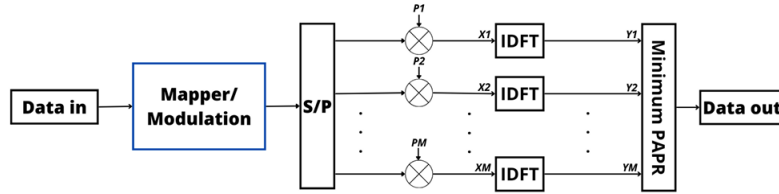


FIGURE 1.7: Block diagram of an OFDM transmitter with Selected Mapping (SLM). U different phase sequences are applied to the frequency-domain data before the IDFT. The candidate signal with the lowest PAPR is selected for transmission.

Interleaving. Multiple interleaving patterns are applied to the data, and the pattern producing the lowest PAPR is selected, analogous to SLM but using permutation rather than phase rotation.

1.3.3 Comparative Summary

TABLE 1.1: Comparison of classical PAPR reduction techniques.

Technique	BER Impact	Complexity	Side Info	Distortion
PTS	None	High	Yes	No
SLM	None	High	Yes	No
Interleaving	None	Low	Yes	No
Clipping	Yes	Low	No	Yes
TI	None	High	No	No
TR	None	High	Yes	No
ACE	None	Moderate	No	No

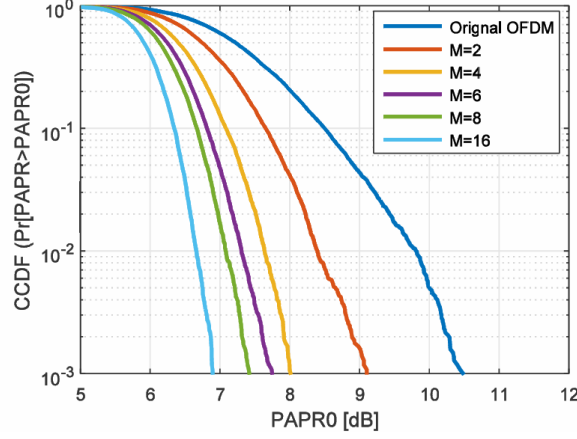


FIGURE 1.8: CCDF of PAPR for conventional SLM with different numbers of phase sequences U ($N = 128$, QPSK). Increasing U improves PAPR reduction but with diminishing returns.

1.4 Limitations of Classical Approaches

Despite their effectiveness, classical PAPR reduction techniques suffer from three fundamental limitations that motivate the machine-learning approaches explored in subsequent chapters.

Computational complexity. Methods such as SLM and PTS require multiple IFFT operations per OFDM symbol. SLM requires U full N -point IFFTs, yielding a per-symbol complexity of $\mathcal{O}(U \cdot N \log_2 N)$, which becomes prohibitive for large N and U .

Side information overhead. SLM and PTS require transmitting $\lceil \log_2 U \rceil$ SI bits per symbol. These bits must be received error-free for correct decoding; an error in the

SI bits causes the entire OFDM symbol to be decoded incorrectly. Protecting SI bits typically requires additional channel coding, further reducing spectral efficiency.

Blind detection difficulty. Eliminating SI overhead through blind detection at the receiver requires U full FFT operations per symbol, and decision metrics become unreliable at low SNR. Blind detection remains an open problem, particularly well-suited to machine-learning-based solutions.

1.5 Summary

This chapter established the OFDM signal model, defined the PAPR problem and its implications for power amplifier efficiency, and surveyed the landscape of classical PAPR reduction techniques. The three core limitations identified — computational complexity, side information overhead, and blind detection difficulty — define the design objectives for the deep learning-based approach presented in the following chapter, where a jointly trained autoencoder replaces the multi-IFFT search with a single neural network forward pass and eliminates the need for side information entirely.

CHAPTER 2

PRNet: PAPR Reduction via Deep Autoencoder

The limitations of classical SLM identified in Chapter 1 — computational complexity, side information overhead, and blind detection difficulty — motivate a fundamentally different approach. Rather than searching over a fixed set of phase sequences, Kim et al. [1] proposed the **PAPR Reduction Network (PRNet)**, the first deep learning-based system for joint PAPR and BER optimisation in OFDM. PRNet treats the transmitter–channel–receiver chain as a single deep autoencoder: the encoder learns a data-dependent constellation mapping that jointly minimises reconstruction error and PAPR, while the decoder learns the corresponding inverse mapping, eliminating the need for side information.

2.1 Autoencoder Framework

2.1.1 Communication System as Autoencoder

In any communication system, the transmitter transforms raw data into a signal suitable for the channel, and the receiver undoes that transformation to recover the original data. This structure maps directly onto the autoencoder paradigm [1]: the encoder corresponds to the transmitter, the bottleneck corresponds to the physical channel (which adds noise and distortion), and the decoder corresponds to the receiver.

The key insight of PRNet is that if both the transmitter and receiver are designed as jointly trained neural networks with the channel model placed between them, the system can be trained end-to-end to minimise any differentiable objective — including PAPR [1].

2.1.2 PRNet Architecture Overview

PRNet inserts two trainable Deep Neural Networks (DNNs) around the standard IFFT/FFT pair [1]. The **encoder** (transmitter-side DNN) replaces the standard constellation mapper: instead of mapping bits to fixed QPSK or QAM points, it produces complex subcarrier values specifically shaped to yield low peaks after the IFFT. The **decoder** (receiver-side DNN) replaces the standard constellation

de-mapper and directly reconstructs the original data from the received noisy frequency-domain subcarriers.

It is important to note that the IFFT and FFT remain fixed, standard mathematical operations within the PRNet pipeline. The encoder replaces the constellation mapper, not the IFFT itself [1].

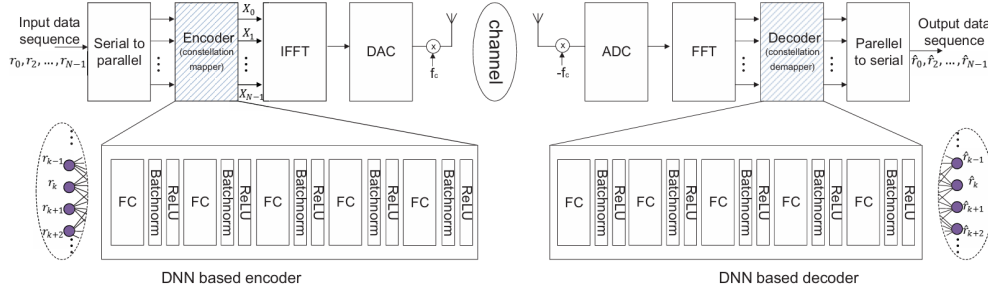


FIGURE 2.1: Structure of the PRNet system with DNN encoder and DNN decoder. Both consist of fully connected (FC) layers with Batch Normalization and ReLU activation. The encoder shapes the constellation before the IFFT to reduce PAPR; the decoder reconstructs the transmitted symbols after the channel and FFT. Reproduced from [1].

Because the encoder and decoder are trained jointly over the full training process, the decoder's weights implicitly encode the complete knowledge of the encoder's mapping. No explicit side information is transmitted between transmitter and receiver, eliminating both the bandwidth overhead and the catastrophic failure mode associated with SI errors in classical SLM [1].

2.1.3 Joint Subcarrier Processing

In standard OFDM with QPSK, the constellation mapping is performed independently per subcarrier: each pair of bits maps to one of four fixed points. PRNet fundamentally changes this by employing a fully connected encoder, where every output subcarrier value is a function of the *entire* input block. This holistic processing enables the encoder to detect input patterns that would produce high PAPR under standard mapping and to compensate by adjusting all subcarrier amplitudes and phases jointly [1].

2.2 System Model

The complete PRNet signal chain can be expressed as [1]:

2.1

$$\hat{\mathbf{r}} = g \circ \text{FFT} \circ \mathbb{H} \circ \text{IFFT} \circ f(\mathbf{r}),$$

where f denotes the encoder, g denotes the decoder, $\mathbb{H}$ represents the channel, and $\mathbf{r}$ and $\hat{\mathbf{r}}$ are the input and reconstructed output vectors, respectively. The following subsections describe each stage.

2.2.1 Input Representation

The system considers $N = 64$ subcarriers with QPSK modulation ($M = 4$ constellation points per subcarrier), yielding $N \times \log_2 M = 128$ bits per OFDM symbol [1]. The symbol index for each subcarrier is $r_k \in \{0, 1, 2, 3\}$, and the full input vector is $\mathbf{r} = [r_0, r_1, \dots, r_{N-1}]^T$.

In the neural network implementation, each symbol index r_k is represented as a one-hot vector of length $M = 4$, and these vectors are concatenated to form the network input [1]:

$$\mathbf{r}_{\text{input}} \in \mathbb{R}^{NM} = \mathbb{R}^{256}.$$

One-hot encoding ensures that the network treats each symbol index as an independent category without imposing a spurious ordinal relationship [1].

2.2.2 Encoder: Learned Constellation Shaping

The encoder is a deep neural network that maps the one-hot input vector $\mathbf{r}_{\text{input}} \in \mathbb{R}^{NM}$ to a vector of $2N = 128$ real numbers representing the real and imaginary parts of N complex subcarrier values [1]:

$$f(\mathbf{r}; \theta_f) \in \mathbb{R}^{2N},$$

where θ_f denotes all learnable weights and biases. The **unstacking operator** $\mathcal{C}(\cdot) : \mathbb{R}^{2N} \rightarrow \mathbb{C}^N$ converts these to complex symbols:

$$\mathbf{X}_{\text{enc}} = \mathcal{C}(f(\mathbf{r}; \theta_f)) \in \mathbb{C}^N,$$

where the k -th encoded symbol is $X_{\text{enc},k} = f(\mathbf{r})_{2k} + j \cdot f(\mathbf{r})_{2k+1}$. These symbols are not constrained to any standard constellation grid; they are continuous-valued complex numbers shaped by the encoder to produce low-PAPR waveforms after the IFFT [1].

2.2.3 OFDM Modulation

The encoded symbols are modulated onto the subcarriers via the standard IFFT [1]:

2.2

$$x[n] = \sum_{k=0}^{N-1} X_{\text{enc},k} e^{j2\pi nk/N}, \quad 0 \leq n \leq N-1.$$

This is the standard IFFT operation; the only difference from conventional OFDM is that the input symbols $X_{\text{enc},k}$ are the encoder's learned values rather than fixed constellation points.

For accurate continuous-time PAPR estimation, the frequency-domain vector can be zero-padded from length N to NL before computing an NL -point IFFT, where L is the oversampling factor. With $N = 64$ and $L = 4$, this produces 256 time-domain samples [1].

2.2.4 Channel Model

After OFDM modulation, the time-domain signal passes through the wireless channel, modelled as [1]:

$$\mathbf{y} = \mathbb{H}(\mathbf{x}) + \epsilon,$$

where $\mathbb{H}$ denotes the channel effects (Rayleigh fading [1]) and ϵ is additive white Gaussian noise (AWGN).

From the neural network's perspective, the channel is a fixed mathematical function within the computation graph. During training, the channel is simulated by generating random noise and fading coefficients. Since these operations (addition, multiplication) are differentiable, gradients from the loss function propagate backward through the channel during backpropagation [1].

2.2.5 Decoder

At the receiver, the FFT recovers the frequency-domain subcarriers:

$$\mathbf{Y} = \text{FFT}(\mathbf{y}) \in \mathbb{C}^N.$$

The decoder DNN processes these received symbols via the **stacking operator** $\mathcal{S}(\cdot) : \mathbb{C}^N \rightarrow \mathbb{R}^{2N}$, which interleaves real and imaginary parts, and produces the reconstructed output [1]:

$$\hat{\mathbf{r}} = g(\mathcal{S}(\mathbf{Y}); \theta_g) \in \mathbb{R}^{NM},$$

where θ_g denotes the decoder's learnable parameters. The output consists of scores for each possible symbol index; the final decision is made by selecting the index with the largest score for each subcarrier.

2.2.6 Signal Dimension Summary

TABLE 2.1: Summary of PRNet signal dimensions for the one-hot formulation [1].

Symbol	Domain / Shape	Meaning
$\mathbf{r}$	$\mathbb{R}^{NM}$ (256)	One-hot encoded input
$f(\mathbf{r})$	$\mathbb{R}^{2N}$ (128)	Encoder output (stacked I/Q)
$\mathbf{X}_{\text{enc}}$	$\mathbb{C}^N$ (64)	Encoded complex subcarrier symbols
$x[n]$	$\mathbb{C}^{NL}$ (64L)	IFFT output (time-domain waveform)
$\mathbf{y}$	$\mathbb{C}^{NL}$ (64L)	Received signal (after channel)
$\mathbf{Y}$	$\mathbb{C}^N$ (64)	FFT output (received subcarriers)
$\hat{\mathbf{r}}$	$\mathbb{R}^{NM}$ (256)	Decoder output (reconstructed scores)

2.3 Network Architecture

Both the encoder and decoder are constructed from repeated sub-blocks, each consisting of a Fully Connected (FC) layer, Batch Normalization (BN), and ReLU activation, applied in sequence [1].

2.3.1 Sub-Block Structure

Each sub-block transforms an input vector $\mathbf{h}^{(\ell-1)}$ into an output vector $\mathbf{h}^{(\ell)}$ through three operations:

Fully Connected Layer. A linear transformation $\mathbf{y}_{\text{FC}} = \mathbf{W}_m \mathbf{h}^{(\ell-1)} + \mathbf{b}_m$, where $\mathbf{W}_m \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and $\mathbf{b}_m \in \mathbb{R}^{d_{\text{out}}}$. The hidden layers use $d_{\text{out}} = 2048$ nodes [1].

Batch Normalization. Normalises the FC output to zero mean and unit variance within each training mini-batch, then applies learnable scale (γ) and shift (β) parameters [1]:

$$|\mathbf{y}_{\text{FC}}|_{\text{norm}} = \gamma \cdot \frac{\mathbf{y}_{\text{FC}} - \mathbb{E}[\mathbf{y}_{\text{FC}}]}{\sqrt{\text{Var}[\mathbf{y}_{\text{FC}}] + \nu}} + \beta,$$

where $\nu = 0.001$ prevents division by zero. BatchNorm stabilises training and allows the use of higher learning rates.

ReLU Activation. The Rectified Linear Unit $\phi(x) = \max(x, 0)$ provides the nonlinearity essential for the network to learn the complex, non-convex constellation shaping required for PAPR reduction [1].

The complete sub-block transformation is:

$$\mathbf{h}^{(\ell)} = \phi\left(\text{BN}\left(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\right)\right), \quad \phi(\cdot) = \text{ReLU}(\cdot).$$

2.3.2 Encoder and Decoder Topology

The encoder consists of $L_f = 5$ FC-BN-ReLU sub-blocks [1]:

2.3

$$f(\mathbf{r}) = \phi_{L_f}\left(\left|\mathbf{W}_{L_f}^f \phi_{L_f-1}\left(\cdots \phi_1\left(\left|\mathbf{W}_1^f \mathbf{r} + \mathbf{b}_1^f\right|_{\text{norm}}\right) \cdots\right) + \mathbf{b}_{L_f}^f\right|_{\text{norm}}\right).$$

The decoder has the same topology ($L_g = 5$ sub-blocks) with its own independent set of weights. The final output layer of each network is a plain FC layer with linear activation (no ReLU), as the outputs must be unconstrained real values [1].

2.4 Training Methodology

2.4.1 Loss Function

PRNet optimises two competing objectives simultaneously: reconstruction accuracy (low BER) and peak suppression (low PAPR). The reconstruction loss measures the Mean Squared Error (MSE) between the original and reconstructed vectors [1]:

2.4

$$\mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) = \frac{1}{NM} \|\mathbf{r} - \hat{\mathbf{r}}\|_2^2 = \frac{1}{NM} \sum_{i=0}^{NM-1} (r_i - \hat{r}_i)^2.$$

The PAPR penalty directly measures the peak-to-average power ratio of the time-domain waveform [1]:

2.5

$$\mathcal{L}_2(\mathbf{r}) = \text{PAPR}\{x[n]\} = \frac{\max_{0 \leq n \leq NL-1} |x[n]|^2}{\frac{1}{NL} \sum_{n=0}^{NL-1} |x[n]|^2}.$$

The joint loss function is a weighted combination [1]:

2.6

$$\mathcal{L}(\mathbf{r}, \hat{\mathbf{r}}) = \mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) + \lambda \mathcal{L}_2(\mathbf{r}),$$

where $\lambda > 0$ controls the trade-off between reconstruction accuracy and PAPR reduction. Larger λ values produce stronger PAPR reduction at the expense of higher BER. The paper adopts $\lambda = 0.01$ [1].

The BER–PAPR Trade-off

The parameter λ governs a fundamental trade-off: more aggressive PAPR reduction requires greater constellation distortion, which degrades the decoder’s ability to distinguish between symbol classes. This trade-off is not a limitation specific to PRNet but a physical constraint of the OFDM system.

2.4.2 Two-Stage Training Procedure

PRNet employs denoising autoencoder training, where noise at corruption level η (noise-to-signal ratio) is injected during training to force the decoder to learn robust decision boundaries [1]:

$$\eta = \frac{\mathbb{E}[|\epsilon|^2]}{\mathbb{E}[|\mathbf{r}|^2]}.$$

Selecting η involves a trade-off: too little noise yields overly tight decision boundaries that fail under real channel conditions, while too much noise prevents the decoder from learning to distinguish adjacent constellation points. Kim et al. employ a two-stage procedure [1]:

1. **Stage 1 — Corruption level selection:** Train multiple models using only the reconstruction loss $\mathcal{L}_1$ ($\lambda = 0$) for a range of η values. Evaluate BER across a full SNR sweep and select the η yielding the lowest overall BER. The paper identifies $\eta^* = -15$ dB [1].
2. **Stage 2 — Joint training:** Fix $\eta = \eta^* = -15$ dB and train the final model with the full joint loss $\mathcal{L} = \mathcal{L}_1 + \lambda \mathcal{L}_2$.

Stage 1 establishes robust reconstruction capability before Stage 2 introduces the PAPR objective, preventing the encoder from distorting the constellation before the decoder has developed adequate noise-handling foundations.

Algorithm 1: Two-stage PRNet training procedure [1].

Input : Candidate corruption levels $\{\eta\}$, PAPR weight λ , steps T , batch size B

Output : Trained parameters (θ_f, θ_g)

```

// Stage 1: select  $\eta^*$  using  $\lambda = 0$ 
1 foreach  $\eta$  in  $\{\eta\}$  do
2   Initialise PRNet parameters  $(\theta_f, \theta_g)$ 
3   for  $t = 1$  to  $T$  do
4     Sample a batch of  $B$  random QPSK symbol indices  $\rightarrow \mathbf{r}$ 
5     Forward pass:  $\hat{\mathbf{r}} \leftarrow \text{PRNet}(\mathbf{r}; \theta_f, \theta_g, \eta)$ 
6     Compute  $\mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}})$  and update  $(\theta_f, \theta_g)$  via Adam
7   end
8   Evaluate BER over an SNR sweep and record result for this  $\eta$ 
9 end
10 Select  $\eta^*$  that minimises BER

// Stage 2: joint training at  $\eta^*$ 
11 Initialise the final PRNet parameters  $(\theta_f, \theta_g)$ 
12 for  $t = 1$  to  $T$  do
13   Sample a batch of  $B$  random QPSK symbol indices  $\rightarrow \mathbf{r}$ 
14   Forward pass: obtain  $\hat{\mathbf{r}}$  and time-domain  $x[n]$ 
15   Compute  $\mathcal{L} = \mathcal{L}_1(\mathbf{r}, \hat{\mathbf{r}}) + \lambda \mathcal{L}_2(x[n])$ 
16   Update  $(\theta_f, \theta_g)$  via Adam on  $\nabla \mathcal{L}$ 
17 end

```

2.4.3 End-to-End Differentiability

Training requires gradients to flow from the loss function through the decoder, FFT, channel, IFFT, and encoder. For the AWGN channel ($\mathbf{y} = \mathbf{x} + \epsilon$), the noise term does not depend on the encoder's parameters, so $\partial \mathcal{L} / \partial \mathbf{x} = \partial \mathcal{L} / \partial \mathbf{y}$. For multiplicative fading channels ($\mathbf{y} = h\mathbf{x} + \epsilon$), the gradient is scaled by the fading coefficient. The IFFT and FFT are linear operations whose gradients are their respective transposes. The max function in the PAPR loss is differentiable almost everywhere, with gradient equal to 1 at the position of the maximum and 0 elsewhere. Consequently, gradients propagate cleanly through the entire system [1].

2.5 Inference and Deployment

2.5.1 Inference Procedure

After training, the encoder and decoder weights are frozen. Inference consists of a single forward pass through the encoder, a standard IFFT for OFDM modulation, transmission over the physical channel, FFT demodulation, and a single forward pass through the decoder. The encoder is a deterministic function: the same input always produces the same output, with no search, candidate comparison, or trial-and-error involved [1].

2.5.2 Generalisation

The encoder is not a lookup table but an algebraic function (composition of matrix multiplications and ReLU operations) whose parameters were optimised to perform well in general. During training, the network processes $500,000 \times 400 = 200$ million random OFDM symbols, a vanishingly small fraction of the $2^{128} \approx 3.4 \times 10^{38}$ possible input sequences. The network nevertheless handles all possible inputs through generalisation: it learns the underlying geometric principles of PAPR avoidance rather than memorising specific input–output pairs [1].

The fully connected architecture produces dense mixing, where each encoded sub-carrier depends on the entire input block. This enables the decoder to exploit a non-standard, globally shaped constellation geometry. However, this property should not be interpreted as a formal frequency-diversity scheme; the system’s robustness to channel effects is limited to the distribution encountered during training [1].

2.6 Simulation Results

2.6.1 Simulation Setup

TABLE 2.2: PRNet simulation parameters [1].

Parameter	Value
Number of subcarriers (N)	64
Modulation	4-QAM/QPSK
Network input representation	$NM = 256$ one-hot entries
Hidden processing blocks per DNN	5
Hidden nodes per layer	2048
Activation function	ReLU (hidden), Linear (output)
Batch normalisation	Yes, after every FC layer
Optimiser	Adam
Learning rate	10^{-4}
Batch size	400
Training steps	500,000
Optimal corruption level (η^*)	-15 dB
PAPR weight (λ)	0.01
CCDF evaluation size	50,000 random OFDM sequences
PTS comparison	4 sub-blocks, 4 phase factors

2.6.2 BER Performance

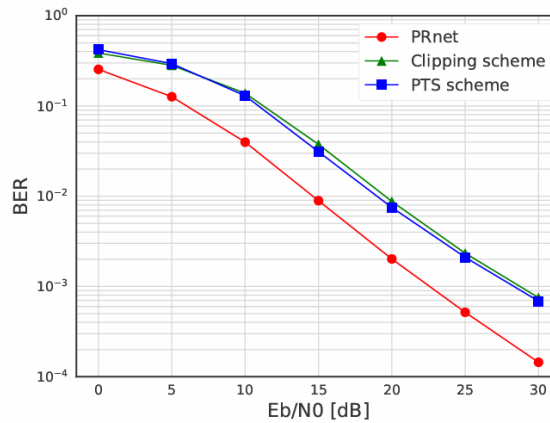


FIGURE 2.2: BER vs. SNR of conventional PAPR reduction schemes and PRNet. PRNet achieves the lowest BER among the plotted schemes because the jointly trained encoder–decoder pair effectively learns a form of implicit joint source–channel coding. Reproduced from [1].

The BER results show that clipping introduces in-band distortion which becomes the dominant error source at high SNR. PTS is distortionless in principle but requires

side information and multiple IFFT computations. PRNet achieves BER lower than the plotted baselines because the encoder shapes the constellation in a way that the jointly trained decoder can exploit, effectively learning an implicit coding gain under the simulated channel [1].

2.6.3 PAPR Reduction

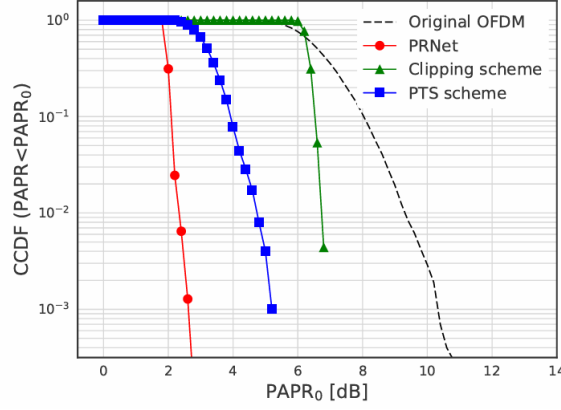


FIGURE 2.3: CCDF of PAPR as reported by the original PRNet paper. The PRNet curve is significantly shifted to the left compared to conventional OFDM. Reproduced from [1].

The CCDF plot demonstrates that PRNet substantially reduces PAPR. The paper reports that the probability of PAPR exceeding 3 dB is less than 0.1% [1], allowing the HPA to operate with substantially less input back-off.

It should be noted that the released implementation computes the CCDF using a peak-amplitude proxy ($10 \log_{10} \max_n |x[n]|$) rather than the standard power-PAPR definition. Standard power-PAPR evaluation yields a PRNet value near 6 dB at $\text{CCDF} = 10^{-3}$, which is still a significant reduction from the conventional OFDM baseline of 10–11 dB, but less extreme than the paper’s visual figure suggests.

2.6.4 Effect of Corruption Level

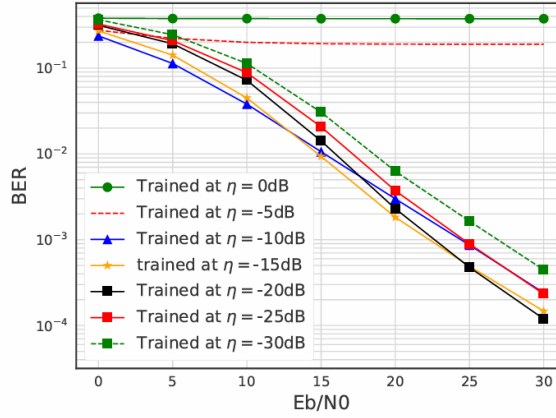


FIGURE 2.4: BER of PRNet trained at different corruption levels η . The model trained at $\eta = -15$ dB consistently yields the lowest BER across the entire SNR range. Reproduced from [1].

The results validate the two-stage training procedure: models trained at very low corruption levels ($\eta = -30$ dB) develop overly tight decision boundaries and fail under real channel noise, while models trained at excessive corruption ($\eta = 0$ dB) cannot learn meaningful boundaries. The optimal corruption level $\eta^* = -15$ dB yields the lowest BER across all tested SNR values [1].

2.6.5 BER–PAPR Trade-off

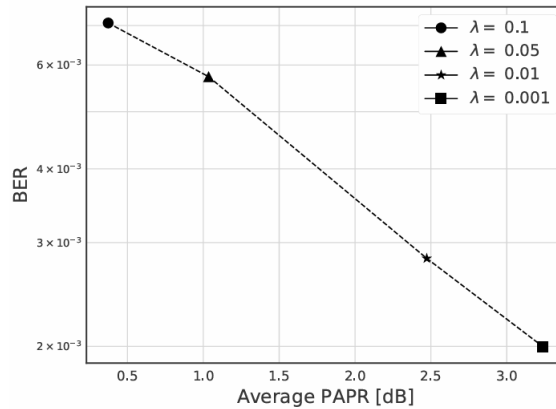


FIGURE 2.5: BER vs. average PAPR by varying λ at SNR = 20 dB. Larger λ reduces PAPR but increases BER, confirming the fundamental trade-off. Reproduced from [1].

As λ increases, PAPR decreases monotonically while BER increases, confirming the fundamental tension between these objectives. At $\lambda = 0.01$, the paper reports a practical operating point with meaningful PAPR reduction and competitive BER [1].

2.6.6 Computational Complexity

At inference time, PRNet replaces the U -fold search of SLM/PTS with a single forward pass. The per-pass complexity is $\mathcal{O}(L \cdot K \cdot M)$, where L is the number of layers, $K = 2048$ is the hidden dimension, and M is the input dimension. This cost is fixed and does not grow with the number of candidates U [1].

TABLE 2.3: Execution time comparison for different PAPR reduction methods [1].

Method	Execution time (μs)
PTS	2,890
Clipping and filtering	1,415
PRNet (without parallelism)	2,304
PRNet (with GPU)	480

With GPU acceleration, PRNet is approximately 6 times faster than PTS and 3 times faster than clipping and filtering, while achieving better BER and PAPR performance [1].

2.7 Limitations

Despite its contributions, PRNet presents several limitations that define the research agenda for subsequent work.

Scalability. The fully connected architecture scales poorly with the number of sub-carriers. The first encoder FC layer has $d_{\text{in}} \times 2048$ weights, where $d_{\text{in}} = NM$ in the one-hot formulation. For $N = 64$ and $M = 4$, this yields 524,288 parameters in the first layer alone. Scaling to practical large-FFT systems would require alternative architectures such as convolutional, unfolding, or attention-based designs [1].

Channel mismatch. Performance is tightly coupled to the channel model used during training. Extending PRNet to frequency-selective channels, imperfect channel state information, and rapidly time-varying scenarios remains an open problem [1].

Configuration dependence. Any change in modulation order (M), number of sub-carriers (N), or pilot structure changes the network’s input/output dimensions or signal statistics, requiring architectural modifications or complete retraining.

PAPR metric ambiguity. The released code computes the PRNet CCDF using a peak-amplitude proxy rather than the standard power-PAPR definition, producing results that appear more favourable than standard evaluation.

Interpretability. The learned constellation shaping is encoded in millions of weight values. Understanding the specific structure the network has discovered, or providing worst-case guarantees, is extremely difficult compared to transparent methods like SLM.

2.8 Related Work

PRNet established the autoencoder paradigm for PAPR reduction. Subsequent works build on this foundation while addressing specific limitations: constrained autoencoders that learn SLM-style phase factors rather than unconstrained nonlinear mappings, signal-cancellation frameworks that achieve PAPR reduction without requiring a jointly trained decoder, and adaptations of the autoencoder architecture to optical OFDM systems with fibre-channel effects [1].

2.9 Summary

This chapter presented PRNet, the first deep learning-based system for joint PAPR and BER optimisation in OFDM. The autoencoder architecture replaces the constellation mapper and de-mapper with jointly trained DNNs, eliminating the multi-IFFT search and side information overhead of classical SLM. The two-stage training procedure identifies the optimal corruption level before introducing the PAPR penalty, and the resulting system achieves both lower BER and lower PAPR than the classical baselines at competitive computational cost. The limitations identified — scalability, channel sensitivity, configuration dependence, and interpretability — define the direction for future work in this area.

APPENDIX A

Simulation Parameters

This appendix consolidates the simulation and system parameters used in the experiments presented in this thesis. Additional parameter tables will be added as further machine learning methods are investigated.

A.1 PRNet Parameters

The following table summarises the parameters adopted for the PRNet experiments, as reported by Kim et al. [\[1\]](#).

TABLE A.1: System and training parameters for PRNet [1].

Parameter	Value
<i>OFDM System</i>	
Number of subcarriers (N)	64
Modulation	QPSK
Oversampling factor (L)	4
Channel (training)	AWGN / Rayleigh flat fading
<i>Network Architecture</i>	
Hidden layers per DNN (encoder/decoder)	5
Neurons per hidden layer	2048
Hidden activation	ReLU
Output activation	Linear
Batch Normalization	Yes (after every FC layer)
<i>Training Configuration</i>	
Optimizer	Adam
Learning rate	10^{-4}
Batch size	400
Training steps	500,000
Optimal corruption level (η^*)	-15 dB
PAPR loss weight (λ)	0.01
<i>Reported Results</i>	
PAPR reduction at CCDF = 10^{-3}	$\approx 6-7$ dB
Inference speedup vs. PTS (GPU)	$\approx 6\times$

APPENDIX B

Glossary of Deep Learning Terms

This appendix provides concise definitions of deep learning concepts referenced throughout the document, intended as a quick reference for readers with a communications background.

Neuron

The basic computational unit of a neural network. It computes a weighted sum of its inputs, adds a bias, and applies an activation function: $y = f(\mathbf{w}^T \mathbf{x} + b)$.

Weight

A learnable parameter in a neural network that scales the connection between two neurons. The collection of all weights (and biases) constitutes the model's trainable parameters, denoted θ .

Bias

A learnable constant added to the weighted sum in a neuron before the activation function is applied: $z = \mathbf{W}\mathbf{x} + \mathbf{b}$. It allows the network to shift the activation function, giving it more flexibility.

Activation Function

A nonlinear function applied element-wise to the output of a neural network layer. Common choices include ReLU, sigmoid, tanh, softmax, and the linear (identity) function. Without nonlinear activations, a multi-layer network would collapse to a single linear transformation.

Fully Connected (FC) Layer

A layer where every neuron is connected to every neuron in the previous layer via a weight matrix $\mathbf{W}$. Also called a dense layer. The computational cost of an FC layer scales as $\mathcal{O}(n_{\text{in}} \times n_{\text{out}})$.

Forward Pass

The computation that propagates input data through the network, layer by layer, to produce an output. During inference (deployment), only the forward pass is needed—backpropagation is not required.

Inference

The process of using a trained model to make predictions on new, unseen data.

In the PAPR context, inference means performing one forward pass per OFDM symbol at real-time rates.

Loss Function

A scalar function that quantifies how far the network's output is from the desired output. Training minimises this function. Common choices include MSE for regression and cross-entropy for classification. In PAPR, custom losses combine PAPR and MSE terms.

Training

The iterative process of adjusting network parameters to minimise the loss function. Involves repeated forward passes, loss computation, backpropagation, and parameter updates over many epochs.

Gradient

The vector of partial derivatives of the loss function with respect to each trainable parameter. It indicates the direction and magnitude of the steepest increase in loss; the optimizer updates parameters in the opposite direction.

Backpropagation

The algorithm used to compute gradients of the loss function with respect to all network parameters by applying the chain rule of calculus, layer by layer, from the output back to the input. These gradients are then used by the optimizer to update the weights.

Optimizer

The algorithm that updates network parameters based on computed gradients. SGD (Stochastic Gradient Descent) is the simplest; Adam (Adaptive Moment Estimation) is the most commonly used in practice, combining momentum and per-parameter learning rate adaptation.

Learning Rate (η)

A hyperparameter that controls the step size of each parameter update. Too large a value causes training to diverge; too small a value makes convergence extremely slow.

Epoch

One complete pass through the entire training dataset. Training typically requires hundreds of epochs before the network converges to a good solution.

Batch

A subset of the training data processed together in one forward and backward pass. Using batches (rather than the entire dataset) enables efficient GPU utilisation and introduces beneficial stochastic noise into training.

Convergence

The point during training at which the loss function stabilises and further epochs yield negligible improvement. A model that fails to converge may be underfitting (too simple) or have a learning rate that is too high.

Hyperparameter

A configuration value set before training begins and not learned from data. Examples include learning rate, batch size, number of layers, number of epochs, and the loss balance factor α .

Overfitting

When a network learns to memorise the training data rather than generalise to unseen data. Symptoms include very low training loss but high validation/test loss. Mitigated by dropout, regularisation, and early stopping.

Regularisation

Any technique that prevents overfitting by constraining the model's complexity. Examples include dropout, weight decay (L_2 regularisation), and batch normalization.

Dropout

A regularisation technique that randomly sets a fraction of neuron outputs to zero during each training step. This prevents neurons from co-adapting and encourages the network to learn redundant representations, reducing overfitting.

Batch Normalization (BatchNorm)

A technique that normalises the inputs to each layer by subtracting the batch mean and dividing by the batch standard deviation. This stabilises training, allows higher learning rates, and acts as a mild regulariser.

Vanishing Gradients

A problem where gradients become extremely small as they propagate through many layers during backpropagation, causing early layers to learn very slowly or not at all. ReLU activations and residual connections help mitigate this.

Autoencoder

A neural network architecture consisting of an encoder that maps input data to a lower-dimensional (or transformed) representation, and a decoder that reconstructs the original data. In the PAPR context, the encoder produces a low-PAPR signal and the decoder recovers the original data symbols.

References

- [1] M. Kim, W. Lee, and D.-H. Cho, “A novel papr reduction scheme for ofdm system based on deep learning,” *IEEE Communications Letters*, vol. 22, no. 3, pp. 510–513, 2017.